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Say the Same, Act Differently

Diagnosing and Steering Text–Action Divergence in Reasoning VLAs

Say the Same, Act Differently: Diagnosing and Steering Text-Action Divergence in Reasoning VLAs

Zihao Feng, Qingzhao Zhang, Chunyu Xia, Bo Yu, Z. Morley Mao, Ramesh Govindan
University of Southern California • University of Arizona • General Motors
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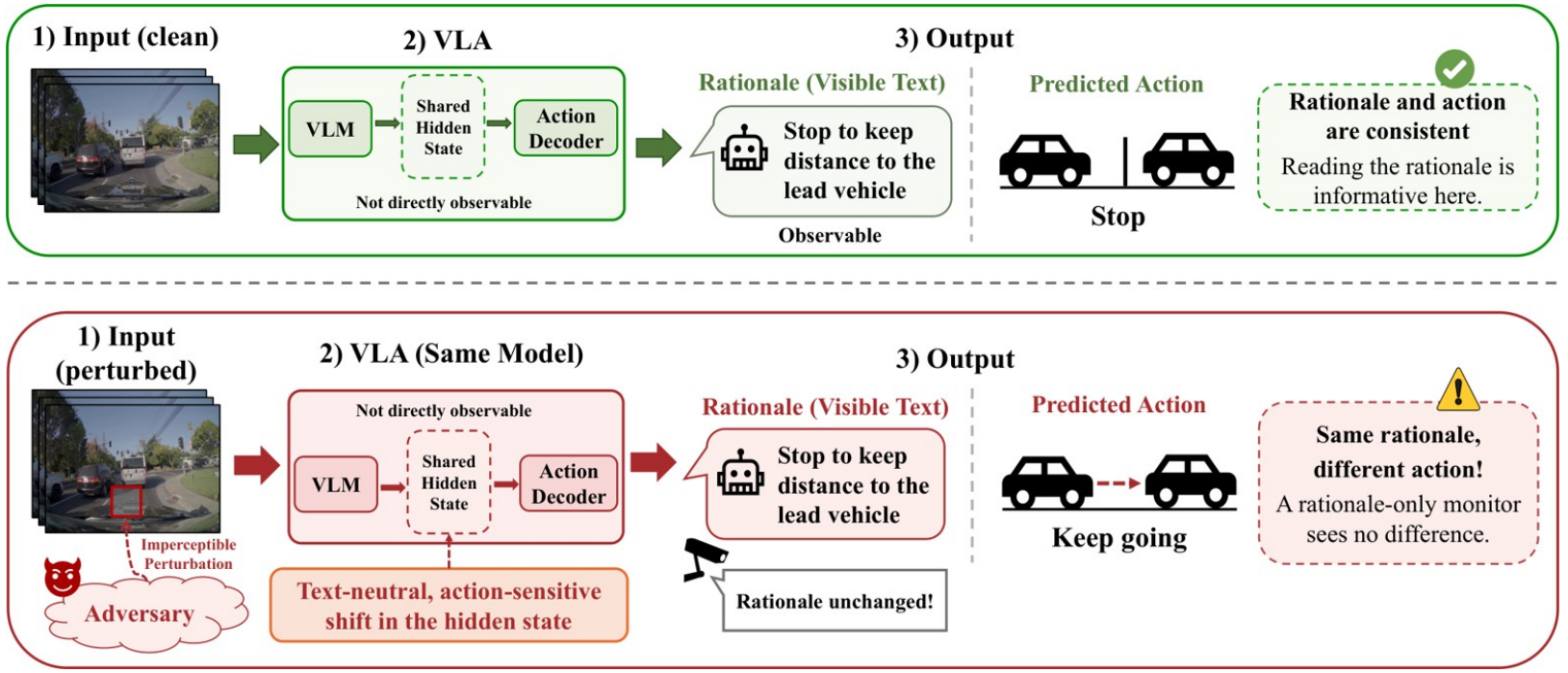
52.6%

Alpamayo driving: identical rationale, yet a new collision (full-image TNAS attack)

Problem

The same rationale can map to opposite actions



Clean input

Says “Stop to keep distance to the lead vehicle” and does stop

Imperceptible perturbation

Identical rationale, but the action becomes Keep going and hits the lead car

Consequence

A rationale-only monitor sees identical evidence in both cases

Prior attacks alter text; we target the unchanged-text blind spot

Prior attacks: alter text → derail action

- TRAP: adversarial patch hijacks CoT
- Trinh et al.: directly tamper with reasoning

→ Tampering leaves visible evidence in the rationale that monitors can catch

This work: action changes, text stays identical

- Visual-input perturbation only, end-to-end backprop
- Driving (Alpamayo-1.5) + manipulation (InstructVLA) embodiments

→ Rationale-only monitors see identical evidence for clean and attacked inputs

Intuition: text is a "low-bandwidth, many-to-one" readout of a shared latent state

Shared VLM latent state
Continuous, high-dim., unobservable

→

Text = discrete readout
Many states → one sentence

→

Action head reads same state
Action-sensitive, text-insensitive DoFs exist

Rationales are informative but coarse: one sentence spans 10–208 m behaviors

84.3%

Predicting "stop or not" from rationale alone AUROC 0.882 (speed only: 0.820)

258

1,181 clips use only 258 distinct rationales

36.9%

of clips' rationales also occur with the opposite stop outcome

245

clips share the most common rationale; forward travel 10.1–208 m

Setup

- Reader: Qwen3-14B (thinking off, greedy decoding), sees only the rationale string, no images, trajectories, speed or maps
- Label: whether Alpamayo plans to stop within a 6.4 s horizon; 1,181 clean clips, one greedy generation each
- Most common: "Keep distance to the lead vehicle since it is directly ahead in our lane"

TNAS: project text gradients out of the top action-driving direction

Hook site (shared latent)

Alpamayo

Reasoning tokens, layers 32–35

InstructVLA

Response tokens, layers 14–17

Intervention: add the same vector δ at each layer over active token span R

Block choice: nonzero text gradient, strong action response; adjacent blocks agree, smaller effect (A.4.1)

1 Action direction d_act

Top right singular vector of action Jacobian J_A

Action-shift norm is non-differentiable at $\delta=0$; Alpamayo uses explicit SVD, InstructVLA random-vector VJP estimates

2 Text-loss subspace $\mathcal{T}$ (rank 2)

Gram–Schmidt(∇NLL, ∇margin)

KL / hinge gradients vanish at $\delta=0$; use target-token NLL and signed margin, still active at the clean point

3 TNAS direction

$d_{TNAS} = \text{normalize}((I - P_{\mathcal{T}})\, d_{act})$

Leaves clean-rationale NLL and margin unchanged to first order, yet still moves the action

Top action-sensitive direction is nearly orthogonal to text gradients

1.87%

Alpamayo: mean ρ = ‖ P_𝒯 d_act ‖ / ‖ d_act ‖ (200 clips, median 1.79%)

5.28%

InstructVLA: mean ρ (200 samples, median 4.90%)

>99.8%

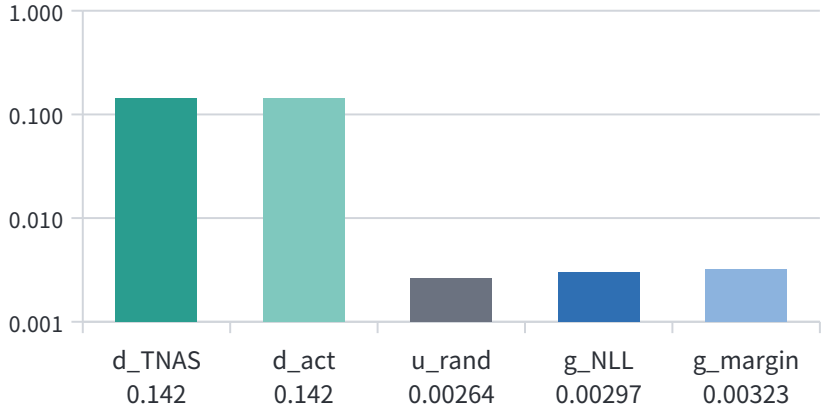
Norm of d_act kept after projecting out text subspace (both models)

Caveat: “nearly orthogonal” alone is of limited informativeness

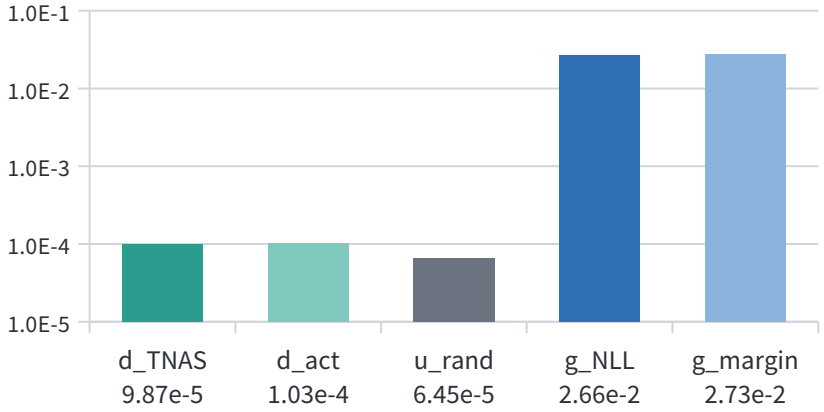
- Random unit vectors in high dimensions have mean ρ **0.978%** (95th pct. 1.91%); d_act is about 2×
- Yet in Alpamayo, **43.5%** of clips exceed the random 95th pct. and **25.5%** the 99th (chance: 5% / 1%)
- TNAS ≈ d_act: the effect mainly arises because the action Jacobian's top direction already lies outside the text subspace → stronger evidence comes from equal-norm interventions

Equal norm: TNAS moves only actions; text gradients move only text

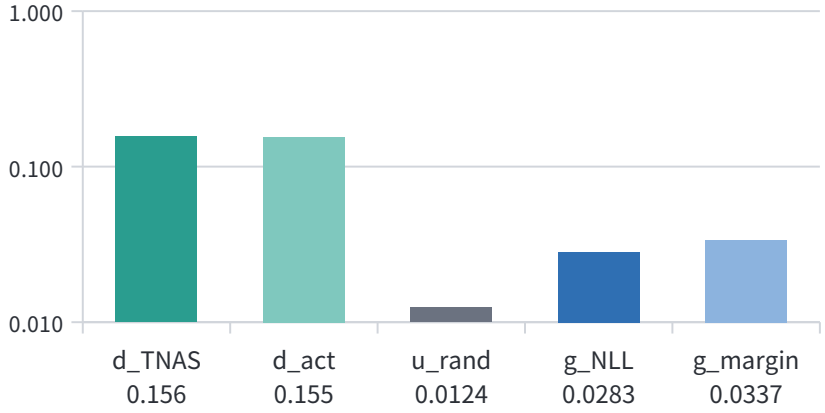
Alpamayo • Action shift ADE (m)



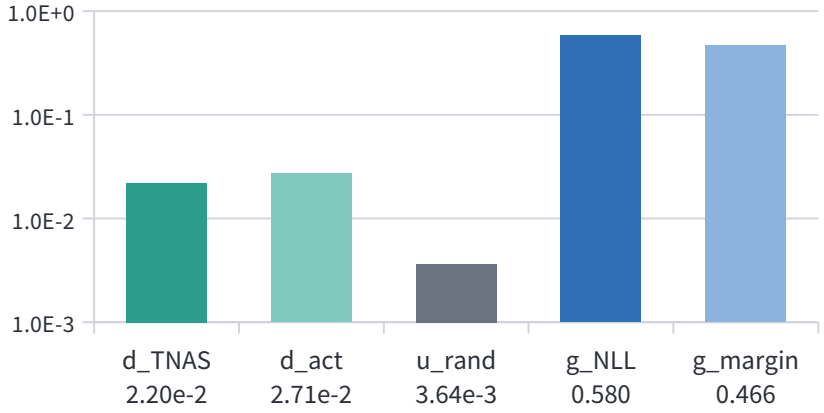
Alpamayo • Text KL



InstructVLA • Normalized action shift



InstructVLA • Text KL



Action shift: TNAS vs. text-gradient direction

44–48×

Alpamayo (vs. random: 54×)

4.6–5.5×

InstructVLA (vs. random: 12.6×)

Text KL: text-gradient direction vs. TNAS

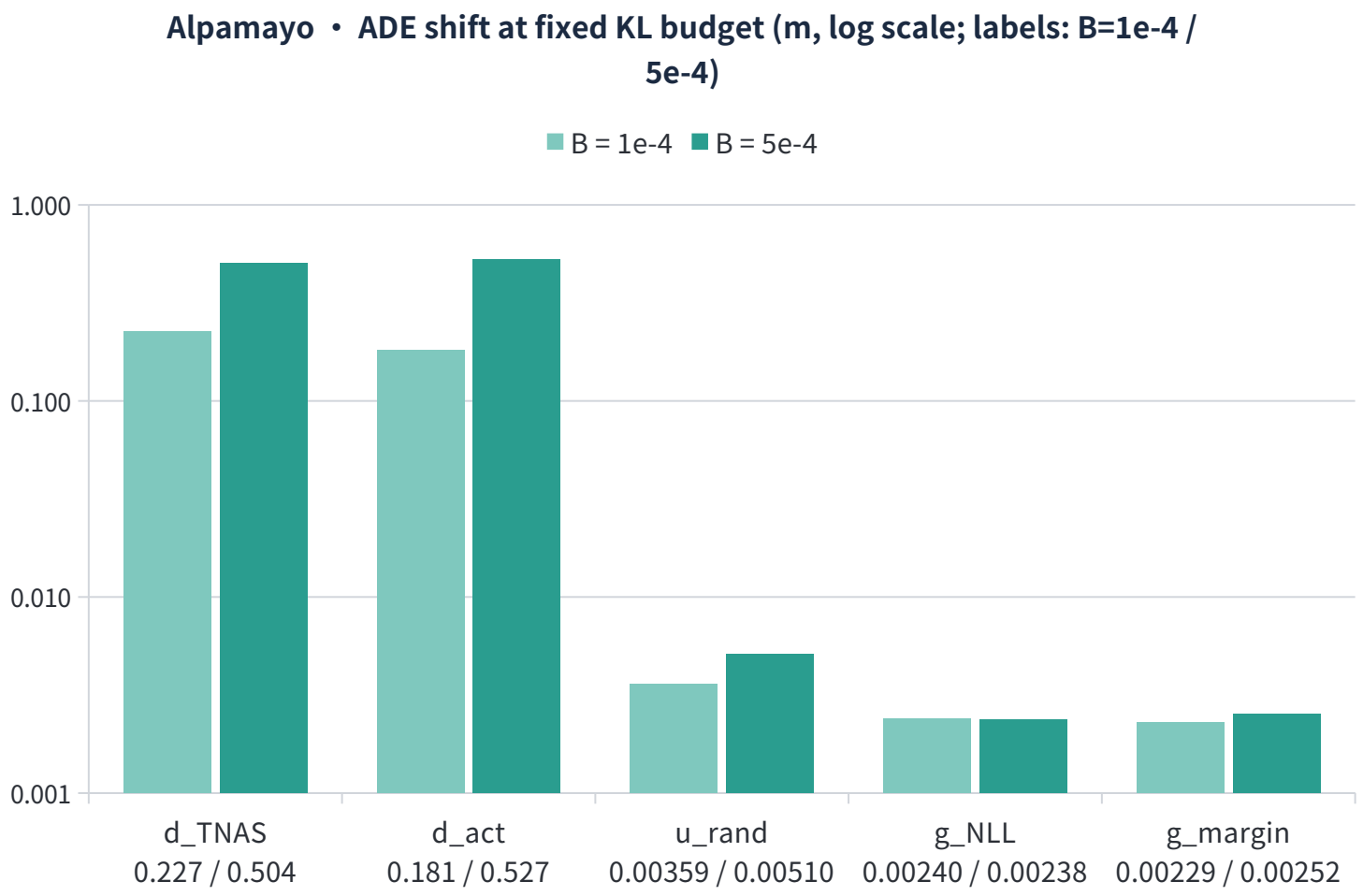
269–277×

Alpamayo

21–26×

InstructVLA

For the same text-drift budget, TNAS moves actions 63–212× more than control directions



63–212×

Alpamayo: TNAS vs. control directions (controls ⩽ 5.10 mm)

InstructVLA （B = 1e-3， n = 200）

Direction	Norm. action shift
d_TNAS	0.0137
d_act	0.0166
u_rand	0.00752
g_NLL	0.00918
g_margin	0.00817

Much smaller advantage on InstructVLA (≈1.5–1.8×)

Attack: optimize in pixel space, steering the latent shift toward TNAS

Threat model

- White-box, digital perturbation
- On normalized image tensor before vision encoder, end-to-end backprop
- 200-step PGD
- Full-image $\varepsilon = 0.05$
- Local patch $\varepsilon = 0.5$ (normalized pixel units)

Eval data: 200 standard clips/samples; collisions on 154 Alpamayo clips with obstacle labels

Action-only PGD

max action shift

Pursues action deviation only

Text-preserving PGD

L_act+txt = action shift − text penalty

Penalizes clean target-token NLL increase and margin drop (Eq. 7)

TNAS-guided PGD

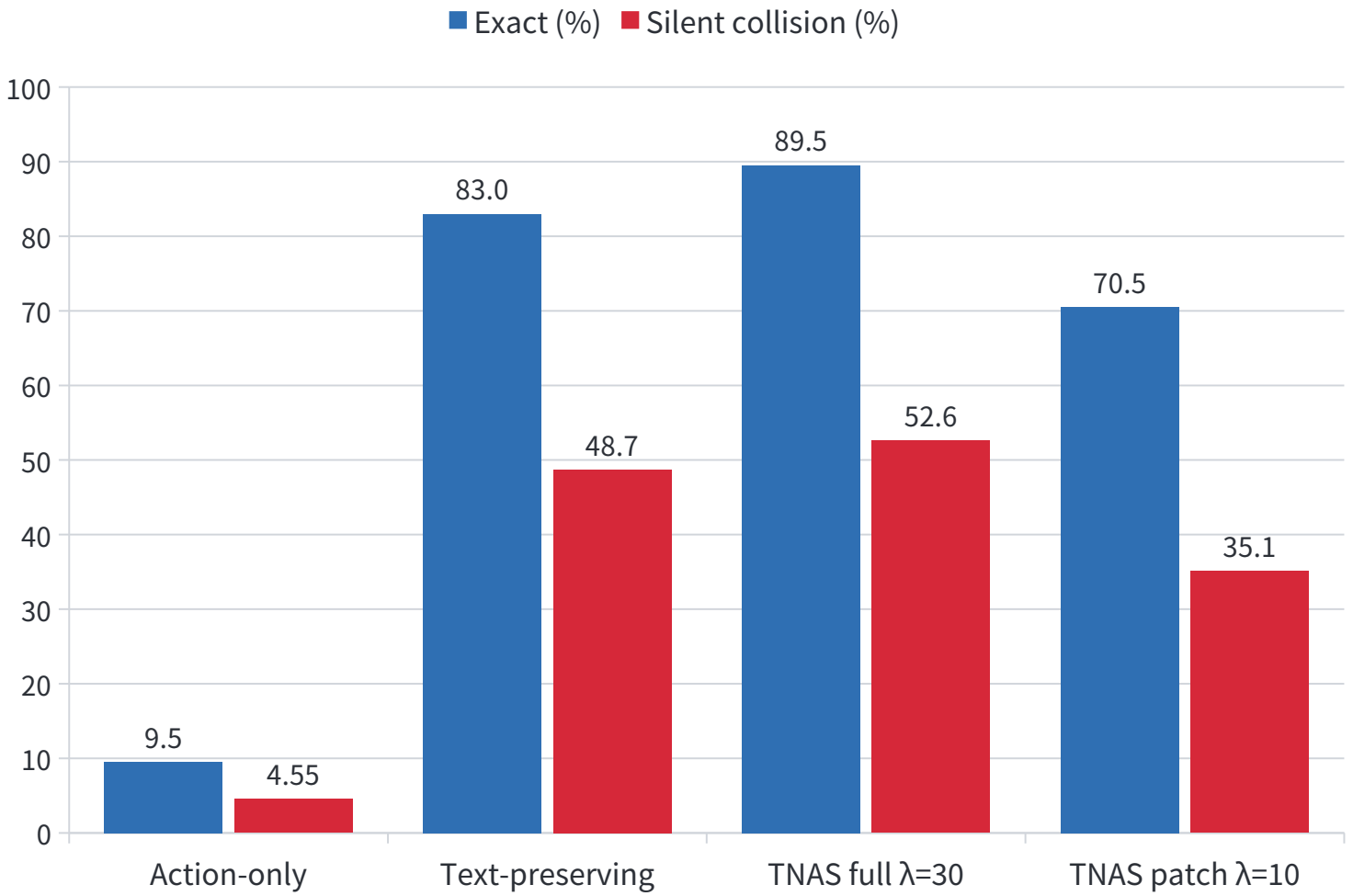
L_act+txt + λ_cos • cos(Δh, d_TNAS)

Also rewards aligning latent shift Δh with d_TNAS (Eq. 8)

Metrics

ADE	KL_tf	Exact	Silent collision
Trajectory shift under fixed prefix (m)	Teacher-forced full-vocab KL	Free-generation exact-match rate	Clean no collision ∧ attacked collision ∧ identical text

Text preservation lifts silent collisions from 4.55% to ~half of scenes

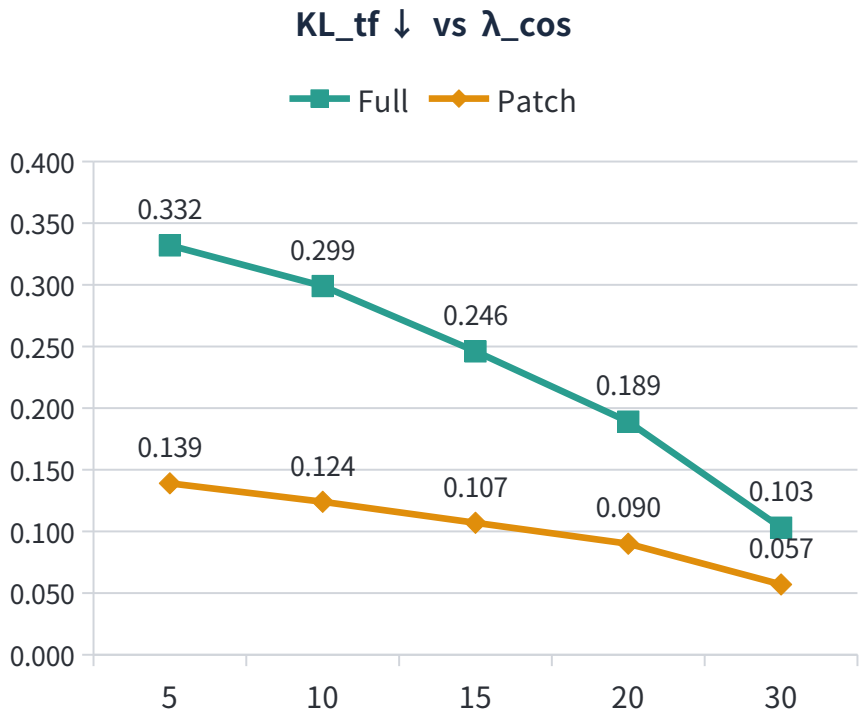


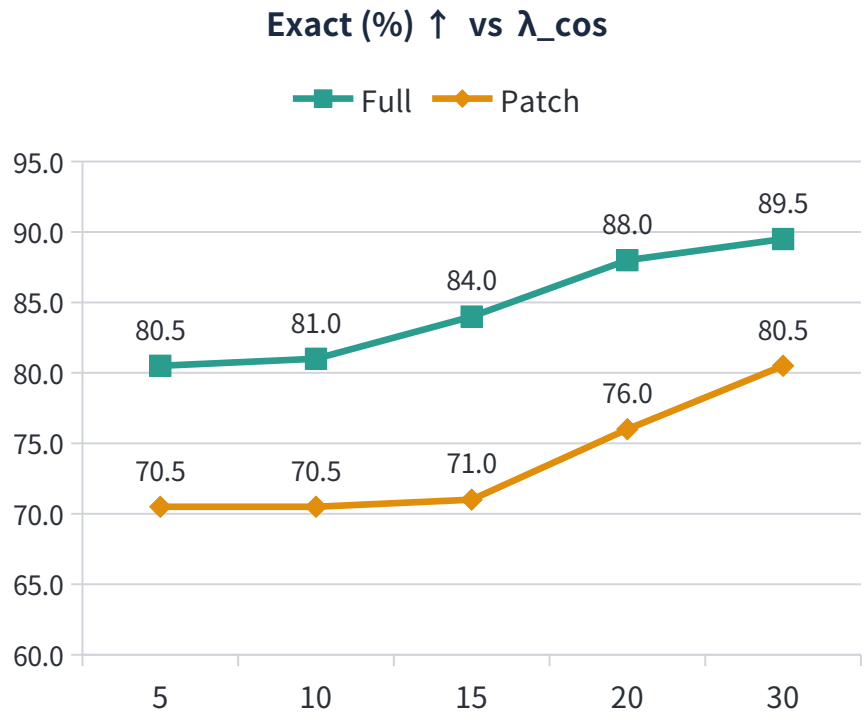
Method	ADE (m)	KL_tf
Action-only	70.1	0.278
Text-preserving	69.2	0.326
TNAS full λ=30	53.0	**0.103**
TNAS patch λ=10	39.8	0.124

52.6%

of 154 obstacle-labeled scenes gain a collision with identical rationale (vs. Action-only 4.55%)

Output-level text penalty fights action gradients; cosine guidance steadily lowers KL, raises Exact





Reference (Text-preserving, λ=0): KL_tf 0.326, Exact 83.0%; x-axis is λ_cos

Full λ=30: KL_tf 68.4% below Text-preserving; patch λ 5→30: KL −59.0%, Exact 70.5% → 80.5%

Conflict

Text penalty raises full-vocab KL

0.278 → 0.326

Penalty only constrains target-token NLL and top-1 margin

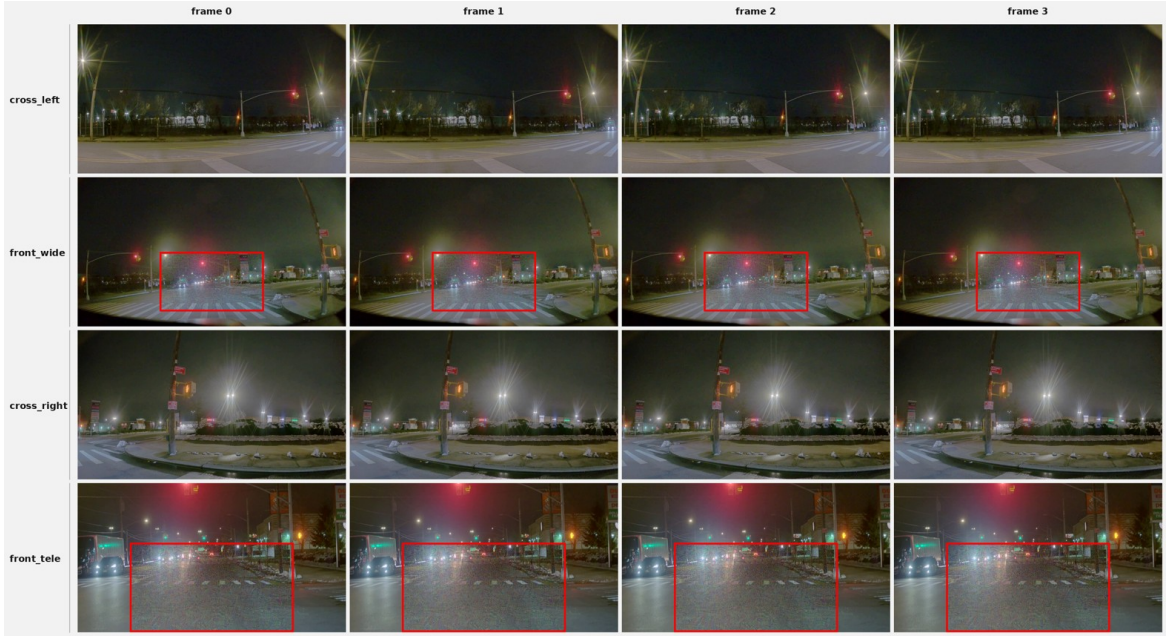
Cost & benefit

ADE 69.2 → 53.0 m, overall collisions flat

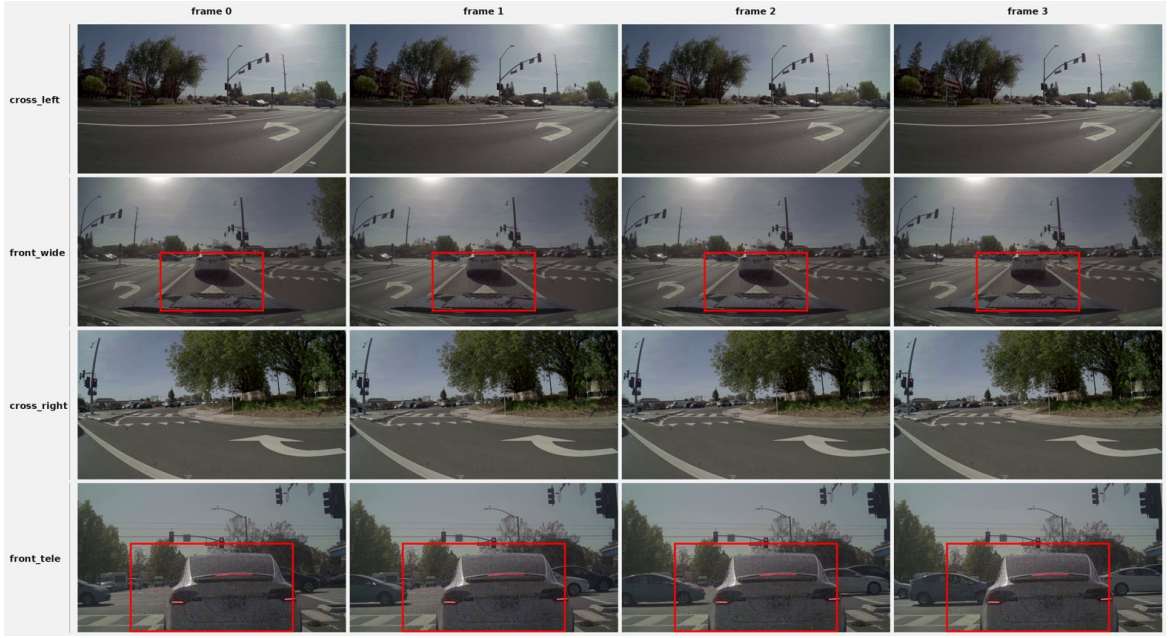
61.7% vs 61.0%

Text fidelity turns into silent collisions

Altering only 13.1% of front-camera visual tokens still yields silent collisions in ~1/3 of scenes



Night intersection: patch near road ahead (rows = 4 cameras, cols = 4 frames)



Day intersection: patch on lead car's rear (red box = patch)

13.1%

visual tokens patched (front tele + rescaled to front wide, shared over 4 frames; side cameras clean)

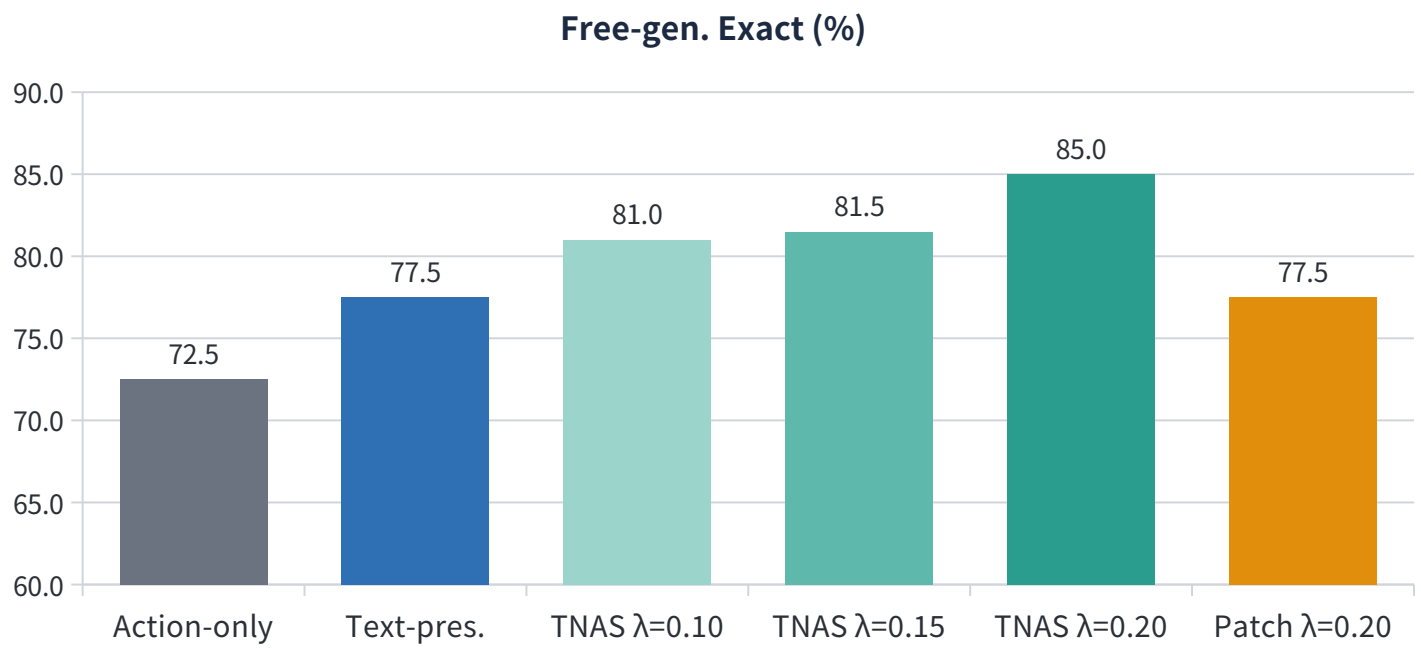
−59%

KL_tf （λ_cos 5→30）， Exact 70.5% → 80.5%

31.8–35.1%

Silent collision rate (154 obstacle-labeled scenes)

Holds for manipulation too: KL −40.5%, Exact reaches 85%



	Action-only	Text-pres.	TNAS λ=0.20	Patch λ=0.20
Action shift (norm.)	2.04	1.97	1.73	1.32
KL_tf	4.66e-2	3.46e-2	**2.06e-2**	2.80e-2

72.5%

responses unchanged under Action-only (no text penalty)

Intents average 9.6 words, 178 strings across 200 samples; one sentence summarizes a 16-step 7-DoF chunk: a very coarse many-to-one bottleneck

TNAS-guided gains

- KL_tf −40.5% (vs. Text-preserving), keeping 87.5% of its action shift
- Removes ~1/3 of residual response changes under Text-preserving
- Patch covers only 16% of image, yet KL is 19.1% below full-image Text-preserving

Exact gains significant; silent-collision gains vs. text-preserving not

Comparison	Metric	Count	p-value	Verdict
Alpamayo： TNAS λ=30 vs Text-preserving	Exact match	179 vs 166	0.0192	Sig.
	Lower KL_tf	179 / 200	1.95e-32	Sig.
	Attack collision	95 vs 94	1.00	Tie
	Silent collision	81 vs 75	0.451	n.s.
Alpamayo： TNAS λ=30 vs Action-only	Silent collision	81 vs 7	2.04e-21	Sig.
InstructVLA： TNAS λ=0.20 vs Text-preserving	Exact match	170 vs 155	0.0135	Sig.
	Lower KL_tf	163 / 200	4.58e-20	Sig.
InstructVLA: patch vs. full Text-preserving	Exact match	155 vs 155	1.00	Tie
	Lower KL_tf	130 / 200	2.65e-5	Sig.

How to read

1. Silent collisions come mainly from adding text preservation: 7 → 75 (Action-only → Text-preserving)

2. TNAS's significant gains are textual: Exact and KL

3. Collisions 81 vs. 75 n.s. → TNAS only marginally boosts attack strength

Key finding: the vulnerability exists

Critical Analysis

Six caveats: strong threat model, open-loop evaluation, limited gains

"Nearly orthogonal" is weak

Random unit vectors: mean ρ 0.978%, d_act only ~2×; TNAS ≈ d_act, so projection does little; key evidence is equal-norm intervention

Limited TNAS gains

Text-preserving already hits 83% Exact, 48.7% silent collisions; TNAS +3.9 pp collisions, n.s.

Shift scale, open loop

ADE 53–70 m; open-loop checks vs. logged agents (non-reactive); no closed-loop or action plausibility eval

Strong threat model

White-box, edits preprocessed tensors; ε=0.5 patch visible, per-clip optimized; no print/physical test

Exact: decoding, granularity

Alpamayo samples (T=0.6, top-p 0.98); InstructVLA intents are coarse, thus easy to keep unchanged

Coverage & defenses

Only two "VLM + flow-matching action head" architectures; no detection/defense evaluated

Summary

Unchanged rationale ≠ safe action: must monitor actions directly

01

New failure mode

First systematic attack with identical rationale but deviated action; end-to-end via vision, in driving and manipulation

02

Mechanism

TNAS: action-sensitive directions nearly orthogonal to text gradients; equal-norm interventions show selectivity gaps of tens to hundreds of times

03

Safety implication

Rationale monitoring (embodied analogue of CoT monitoring) alone cannot certify safety; pair it with action checks and runtime verification

In one sentence: text is a low-bandwidth, many-to-one readout of the latent state, while actions exploit degrees of freedom text cannot see.

In Alpamayo, text preservation alone yields identical-rationale collisions in 48.7% of scenes; 52.6% with TNAS guidance.